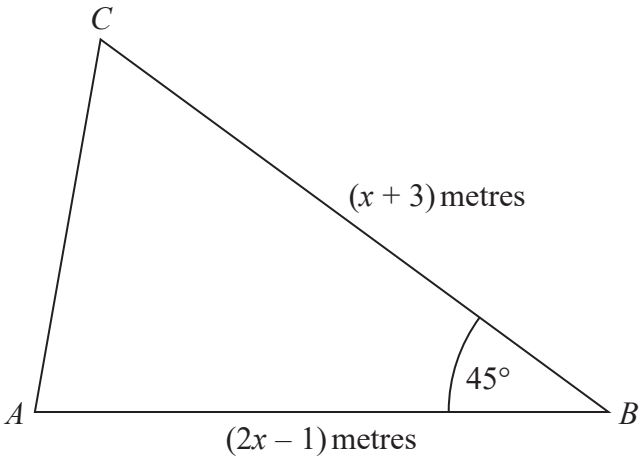
15



The area of triangle ABC is $6\sqrt{2}$ m^2.

Calculate the value of x.
Give your answer correct to 3 significant figures.

Area = ½ab sin C:

$\frac{1}{2}(x + 3)(2x - 1)\sin 45° = 6\sqrt{2}$

1 mark
Use the triangle area formula

1 mark
Set area equal to 6√2

×	$2x$	-1
x	$2x^2$	$-x$
$+3$	$+6x$	-3

$\sin 45° = \frac{\sqrt{2}}{2}$

So $(x + 3)(2x - 1) = 24$

$2x^2 + 5x - 3 = 24$

$2x^2 + 5x - 27 = 0$

1 mark
Rearrange to a quadratic

$x = \frac{-5 \pm \sqrt{(5^2 - 4 \times 2 \times (-27))}}{2 \times 2}$

1 mark
Use the quadratic formula

$x = 2.631043...$ or $x = -5.131043...$

Reject the negative value: lengths must be positive.

2.63

1 mark
x = 2.63 to 3 significant figures

(Total for Question 15 is 5 marks)